

ADHAM ELSAYED

AI Engineer | Generative AI, Machine Learning & Computer Vision

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PROFESSIONAL SUMMARY

Final-year AI Engineering student building complete AI systems across generative AI/RAG, machine learning, computer vision, and evaluation. Delivered evidence-grounded RAG platforms, YOLO/ONNX edge deployments, and Power BI analytics across healthcare, education, retail, and workforce domains. 2nd Place - AI Hackathon Mansoura 2026.

EDUCATION

Mansoura University

B.Sc. in Artificial Intelligence Engineering | GPA: 3.13/4.00 | Grade: B+

Sep 2023 - Expected Feb 2027

Dakahlia, Egypt

TECHNICAL SKILLS

Generative AI & LLM Engineering: RAG, LLM APIs, Prompt Engineering, Grounded Generation, BGE Embeddings, ChromaDB, BM25, RRF, NLI Verification, Hugging Face

AI Evaluation & Safety: Precision@K, Hit@K, MRR, Citation Validation, Claim Verification, Safety Guardrails, Fail-Closed Fallbacks

Machine Learning & Computer Vision: PyTorch, TensorFlow, YOLO, OpenCV, Deep Learning, CNNs, NLP, Object Detection

Backend, Data & BI: Python, FastAPI, REST APIs, SQL, Pandas, Power BI, DAX, MLflow

Deployment, Edge & Tools: Docker, Git, GitHub, Azure AI, Raspberry Pi 5, ONNX Runtime, Firebase, Flutter, Next.js, Arduino, C++

PROJECTS

MedFlow - Evidence-Grounded Clinical RAG | *RAG, BGE, ChromaDB, BM25, RRF, FastAPI, Groq, Python* | **2nd Place - AI Hackathon Mansoura 2026** [GitHub](#)

Architected a hybrid dense+sparse retrieval pipeline (BGE, BM25, RRF) with evidence gating, driving controlled ANSWER / CAUTION / ABSTAIN / REDIRECT behavior.

Benchmarked chunking/Top-K configurations to 200-token/Top-K=4 retrieval: 53.12% Precision@4, 87.50% Hit@4, ~0.7031 MRR.

Validated grounded generation on 7/7 cases: 100% citation validity, metadata resolution, claim coverage, zero fabricated citations.

MedicalPlab - Adaptive Evidence-Grounded Medical Learning | *Generative AI, RAG, NLI, FastAPI, Next.js, Three.js* [GitHub](#)

Designed an adaptive tutoring loop with learner-state tracking, distractor-level signals, and a bounded 3-turn Socratic flow (probe → guide → consolidate), verified via transfer assessment.

Assembled a shared evidence engine with hybrid BM25+dense retrieval, RRF, and cross-encoder reranking, with provenance gating and NLI verification (PMCID/DOI).

Separated generative tutoring from deterministic safety: SAFE_FALLBACK withholds unsupported content, keeping anatomy scoring and exam logic verified and non-generative.

Smart Basket - Computer Vision & Edge AI | *YOLO, OpenCV, Raspberry Pi 5, ONNX, Firebase, Flutter, Python* [GitHub](#)

Trained a YOLO detector (4-class dataset) reaching 96.89% mAP@0.5 / 84.79% mAP@0.5:0.95, then deployed ONNX edge inference on Raspberry Pi 5, syncing cart state to a Flutter app via Firebase.

Code AI Proctor - AI Exam Monitoring Prototype | *YOLO, PyTorch, FastAPI, Azure SQL, Docker, Python* [GitHub](#)

Prototyped webcam-based exam monitoring (YOLO inference, threshold alerts, auditable FastAPI workflow) secured with API-key authentication, optional Azure SQL persistence, and Docker packaging.

EXPERIENCE

- Digitera Bootcamp (YIEP 2026) | EraaSoft x iCareer**
AI Track Participant - Digitera C01

Sep 2026
Egypt

Joined the AI track of a career-focused bootcamp centered on workplace simulation and industry-focused preparation.
- Data Analysis Intern | VOLTIX**
Data Analytics / Business Intelligence

2026

Cleaned and validated data for 1,470 employees, delivering a Power BI HR Attrition dashboard (DAX KPIs) on overtime, tenure, compensation, and satisfaction.
- AI Hackathons Program Participant | Orange Digital Center Egypt x CREATIVA Innovation Hubs**
Applied AI / Rapid Prototyping Program

Aug 2026
Egypt

Prototyped and presented an applied AI solution under a fixed hackathon timeline, sharpening rapid iteration and communication.
- Digital Egypt Pioneers Initiative (DEPI)**
Machine Learning Engineering Trainee - Microsoft Machine Learning Engineer Track, AI & Data Science

Sep 2025 - Jul 2026
Egypt - Remote

Completed a 10-month program covering preprocessing, model training, evaluation, deep learning, NLP, computer vision, Azure AI, MLOps, MLflow, and Hugging Face.
- Robotics Software Engineering Trainee | ST Smart**
Robotics & AI Training

Jul 2025 - Sep 2025
Mansoura, Egypt - Onsite

Completed 50+ hours of robotics training (sensors, motor control, C++/Arduino) and tested 20+ simulation scenarios in object tracking and real-time decisions.

HONORS & AWARDS

- 2nd Place - AI Hackathon Mansoura 2026**
Team MedFlow | Orange Digital Center Egypt & CREATIVA Innovation Hubs

Aug 2026
Mansoura, Egypt

CERTIFICATIONS

- **Building with the Claude API** | Anthropic - Claude Academy
- **Generative AI Chat App with Microsoft Foundry; Intro to Generative AI and Agents** | Microsoft
- **Intro to Modern AI; Intro to Data Science; Python Essentials 1** | Cisco Networking Academy
- **Exploring Artificial Intelligence** | IBM SkillsBuild
- **One Million Prompters - Prompt Engineering** | Dubai Future Foundation
- **HCIA-AI Course** | Huawei